Approximate Computing		
Paderborn University		SoSe 2026
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Lecturers: **Dr. Muhammad Awais** **Prof. Dr. Marco Platzner**		**Lab Module 2:** **Tutorial 2**

1 Objective

Get familiar with Synopsys Design Compiler (DC).

2 Before Start

Before starting, be sure that you have access to the computers in the pool room with your IMT account. If not, please go to:

`https://serviceportal.uni-paderborn.de/`

select **"User Settings > Additional Services"**, and activate your access to **"Computer Engineering Group pool room"**. Moreover, be sure that under the same settings, you have activated your access to **PC2 (Paderborn Center for Parallel Computing)**.

3 VHDL Simulator

You may use Questasim for simulating your designs in this tutorial. Alternatively, Xilinx (now AMD) ISE/Vivado can be used (if you are familiar with it and prefer to use it).

4 Introduction to Logic Synthesis

Normally, a design has to respect a set of specifications (parameters are for instance: processing performance, area, power, etc.). Since the synthesis tool does not know any of these specifications, they have to be provided to the tool. This is done by constraining the design before the compilation process. The constraints define your "wishes" to the compiler. The compiler will do its best to map your HDL description into a netlist that matches the constraints. However, there is no guarantee that these constraints will be met. You will have to check after the

compilation if the circuit is as small/fast as you like. Sometimes the synthesizer will not be able to meet all the constraints, resulting in a larger or slower circuit.

4.1 Optional: Access Synopsys DC from your computer remotely

We have also set up a virtual machine with Synopsys DC set up on it. You can access it via the following steps

1. Send an access request to `mawais@mail.upb.de` with subject line **"Access AxC26 VM"** and your IMT username in the body. Note that this is a one-time request as both QuestaSim and Synopsys Design Compiler are available on the same machine, so if you have already requested access, no further action is needed.
2. Make sure you are connected to the university network either physically or via the VPN.
3. Use *ssh* to connect to the virtual machine: ssh *your_imt_username*@ceg-axc.cs.upb.de
4. You will land in your user local directory where you can store your files. In order to access Synopsys Design Compiler, use the following command:

```
$ singularity exec /ceg/container/Synopsys_2015.06-SP4.sif dc_shell
```

5 Exercise 1: 8-bit Decoder

In this section, we will go through the synthesis of a small design step by step and explore different possibilities at certain points. Download the tutorial archive (tutorial-dc.tar.gz) from PANDA and extract it:

```
tar xvzf tutorial-dc.tar.gz
```

Now, we will investigate how we can use constraints to change the way a combinational circuit is synthesized. The functionality of the design we will use for our investigation is given in: `/tutorial-dc/sourcecode/decoder8.vhd` which is an 8-bit decoder.

Have a look at the source code in order to understand it using Questasim. Open the file decoder8.vhd

- compile
- simulate the design
- view > wave
- drag and drop signals you want to observe from the object window to the wave window
- check the functionality for some arbitrary values

Now, we want to use Synopsys Design Compiler to synthesize the circuit. For this purpose, please follow the following steps:

1. Run the Synopsys Design Compiler:

```
cd tutorial-dc
singularity exec /ceg/container/Synopsys_2015.06-SP4.sif dc_shell
```

2. Set up the library (by issuing the following commands):

```
set target_library \{./LIB/cmos22nm.db}
set link_library \{./LIB/cmos22nm.db}
```

3. Analyze and elaborate the design (by issuing the following commands):

```
analyze -f vhdl ./sourcecode/decoder8.vhd
elaborate decoder8
```

Optional: Additionally, a graphical user interface can be activated by the following command (useful if you want to view the generated hardware).

```
start_gui
```

However, if you are running DC on a remote machine, you might need to include additional switches in your `ssh` command.

4. Compile and see what will be generated by Synopsys:

```
compile
```

Notice that the compilation process reports some information about the design. More detailed reports can be obtained by means of Synopsys report commands

5. QoR reporting:

```
report_qor
```

6 Exercise 2: ALU

As a second example, we will investigate how we can use constraints to synthesize a larger combinational circuit. Look at /tutorial-dc/sourcecode/alu.vhd and try to understand the VHDL code functionality.

```
dc_shell -f scripts/script_alu.tcl
```

check the script alu.tcl and try to understand it.

The following command specifies that the maximum delay to all outputs is limited to 0.8 ns

```
set_max_delay 0.8 -to [all_outputs]
compile
report_area
report_power
```

- Change the max delay to 0.5 ns, 0.3 ns, and 0.1 ns and then discuss about the result of synthesis process.

6.1 Setting output load

The propagation delay of a CMOS circuit depends also on the capacitive load it is driving at the outputs. Basically, the more capacitance a gate has to drive, the longer it will take to charge (or discharge) the load capacitance.

```
set_load [expr 0.002] [all_outputs]
compile
report_timing
report_area
report_power
```